MIL-E-1/1131D
24 June 1983
SUPERSEDING
MIL-E-1/1131C
13 September 1974

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6877

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Triode, miniature, low-Mu

Outline	---	6-4 (EIA)
Base	---	E9-1
Envelope	---	T6-1/2
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9
Element	---	g	nc	h	a	h	g	nc	k	a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Ehk	Rg	Ik	Ic	Pp	TE	Alt
Unit:	V	V dc	V dc	V	Meg	mA dc	mA dc	W	°C	ft
Maximum:	6.6	200	5, -150	450	1	100	5	12	300	Note 1
Minimum:	6.0	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	100	-12	---	---	---	---	---	---	---

GENERAL:

Qualification - Not required

6877
FSC 5960

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits		Unit
							Min	Max	
	Quality conformance inspection, part 1								
1256	Electrode current (1) (anode)	2		0.4	II	Ib	60	90	mA dc
1256	Electrode current (2) (anode)		Ec = -100 V dc	0.4	II	Ib	---	2.0	mA dc
1266	Total grid current	2		0.4	II	Ic	0	-0.5	µA dc
1301	Heater current			0.4	II	If	740	860	mA
1306	Transconductance (1)			0.4	II	Sm	5,500	7,500	µmhos
1336	Heater-cathode leakage		Ehk = 450 V dc	0.4	II	Ihk	---	25	µA dc
1201	Short and discontinuity detection			0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes		E(g-all) = -200 V dc E(a-all) = -300 V dc	2.5	S3	R	50	---	Meg
1031	Sweep-frequency vibration	3	Rp = 2,000 ohms; Ec = -20 V dc; 2.5 G	6.5	Code H	Ep	---	750	mV ac
1266	Grid emission	4		2.5	I	Ic	0	-1.0	µA dc
1306	Transconductance (2)		Ef = 5.7 V	2.5	I	ΔSm Ef	---	10	%
1316	Amplification factor			2.5	I	Mu	3.0	4.5	---
1041	Shock	5	450 G; Ehk = +450 V dc	6.5	See note 6	---	---	---	---
1031	Vibration-fatigue	3		6.5	See note 6	---	---	---	---
---	Shock and vibration-fatigue-test end points								
1031	Sweep-frequency vibration	3	Rp = 2,000 ohms; Ec = -20 V dc; 2.5 G	---	---	Ep	---	1,000	mV ac
1336	Heater-cathode leakage		Ehk = 450 V dc	---	---	Ihk	---	50	µA dc
1306	Change in transconductance (1) of individual tubes			---	---	ΔSm t	---	15	%
1266	Total grid current			---	---	Ic	0	-1.0	µA dc

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits		Unit
							Min	Max	
	Quality conformance inspection, part 2 - Continued								
1121	Base strain			---	---	---	---	---	---
2126	Envelope strain			2.5	I	---	---	---	---
1105	Permanence of marking			---	---	---	---	---	---

Method	Requirement or test	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
	Quality conformance inspection, part 3						
1506	Heater-cycling life		Ef = 7.5 V; Ehk = 450 V dc (cathode positive); Ec = Eb = 0; 1 minute "on", 4 minutes "off"	---	---	---	---
---	Heater-cycling life-test end point:						
1336	Heater-cathode leakage		Ehk = 450 V dc	Ihk	---	50	µA dc
1516	Stability life		Ec = 0; Eb = 150 V dc; Rk = 250 ohms; Ehk = 450 V dc (cathode positive); Rg = 0.5 Meg; TA = room	---	---	---	---
---	Stability life-test end point: (1 hour)						
1306	Change in transconductance (1) of individual tubes			ΔSm_t	---	15	%
1501	Intermittent life	7,8	Ec = 0; Eb = 150 V dc; Rk = 250 ohms; Ehk = 450 V dc (cathode positive); Rg = 0.5 Meg; TE = 300°C	---	---	---	---
---	Intermittent life-test end points (500 hours):	9					
1266	Total grid current			Ic	0	-1.0	µA dc
1301	Heater current			If	720	880	mA
1306	Change in transconductance (1) of individual tubes			ΔSm_t	---	15	%

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Method	Requirement or test	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
	Quality conformance inspection, part 3 - Continued						
---	Intermittent life-test end points (500 hours): -Continued	9					
1306	Transconductance (2)		Ef = 5.7 V	$\frac{\Delta Sm}{Ef}$	---	10	%
1336	Heater-cathode leakage		Ehk = 450 V dc	Ihk	---	50	µA dc
1211	Insulation of electrodes		E (g-all) = -200 V dc E (a-all) = -300 V dc	R	50	---	Meg
1306	Transconductance (1), average change			Avg $\frac{\Delta Sm}{t}$	---	10	%
---	Intermittent life-test end points (1,000 hours)	9					
1266	Total grid current			Ic	0	-1.0	µA dc
1301	Heater current			If	720	880	mA
1306	Change in transconduct-ance (1) of individual tubes			$\frac{\Delta Sm}{t}$	---	15	%
1336	Heater-cathode leakage		Ehk = 450 V dc	Ihk	---	50	µA dc
1211	Insulation of electrodes		E (g-all) = -200 V dc E (a-all) = -300 V dc	R	50	---	Meg

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. This test shall be performed at the conclusion of the holding period.
3. The TUT shall be swept continuously during a time of approximately 1 minute through the frequency range of 50 to 500 Hz and back to 50 Hz. In each mounting plane, the tubes shall experience at least one complete frequency sweep.
4. Prior to this test, the tubes shall be preheated a minimum of 5 minutes operating at the conditions indicated below. The 3-minute test shall not be permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid emission test.

Ef	Ec	Eb
V	V dc	V dc
7.5	-12	100

5. A grid resistor of 0.1 megohm shall be added, except when a thyratron-type short indicator is used.
6. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code E, shall apply.

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7. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a TUT having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

8. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples and shall not exceed a total of three tube failures.

9. The allowable defectives per characteristic shall not be greater than 1 for the first sample and shall not be greater than 3 for combined samples.

10. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 99
DLA - ES

User activities:
Army - AR
Navy - AS, CG, MC, OS, SH

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5960-3300)

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